

Hackathon - 2025

Machine Learning & Artificial Intelligence

Presented by:

Showkat Ahmad Dar

Team 5 NIA/NIH

National Institutes of Health



What is Intelligence and Artificial intelligence?

Intelligence:

The ability to learn, understand and think in a logical way about things

Artificial intelligence:

The study and development of computer systems that can copy intelligent human behaviors.

(Def. Oxford Dictionary)

Why Artificial Intelligence ?



Iron Man



Terminator

Why Artificial Intelligence ?

Just as electricity transformed almost everything 100 years ago, today I actually have a hard time thinking of an industry that I don't think AI will transform in the next several years.

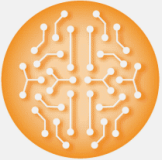


[Andrew Ng, co-founder Coursera](#)

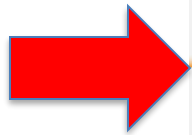
Can we think of some examples ?

Levels of AI?

Three types of Artificial Intelligence

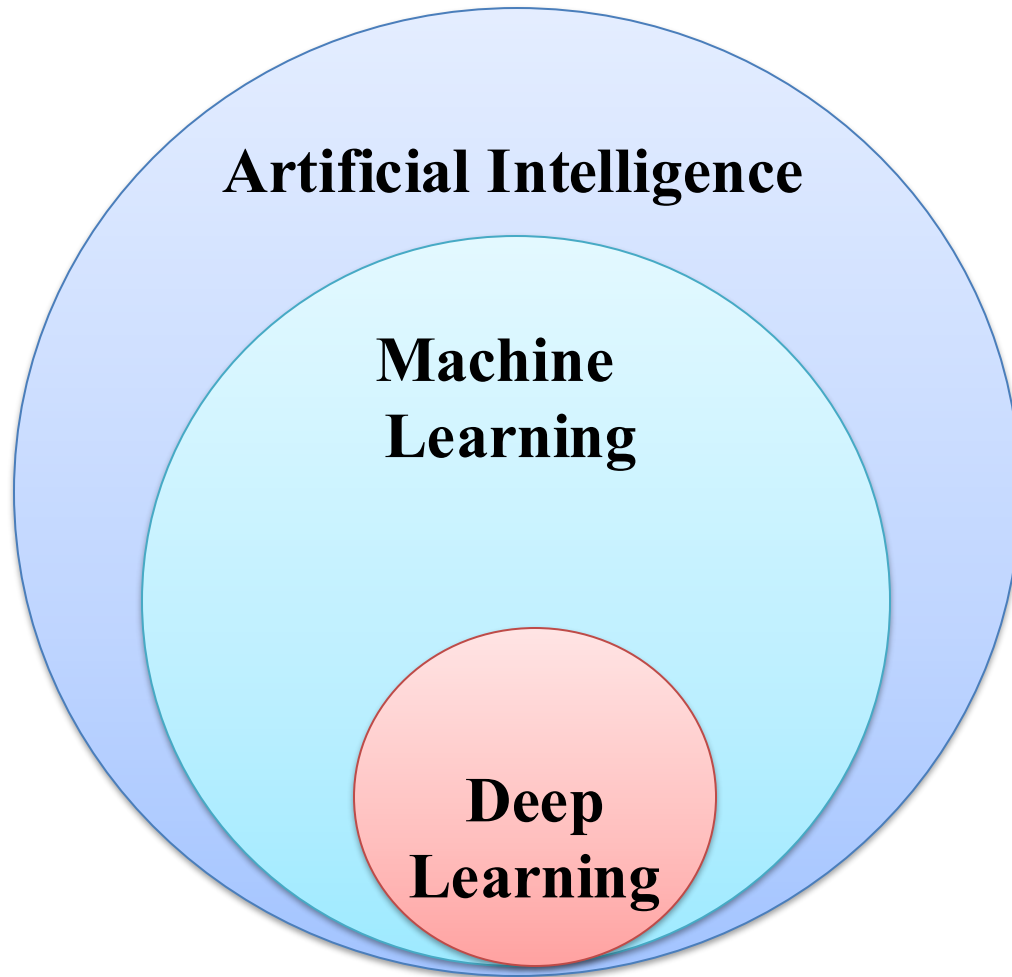
	Artificial Narrow Intelligence (ANI)	Stage-1	Machine Learning	Specialises in one area and solves one problem
				
	Artificial General Intelligence (AGI)	Stage-2	Machine Intelligence	Refers to a computer that is as smart as a human across the board
				
	Artificial Super Intelligence (ASI)	Stage-3	Machine Consciousness	An intellect that is much smarter than the best human brains in practically every field

We are here



Source <https://deltalogix.blog/>

Artificial Intelligence Vs Machine learning Vs Deep Learning



ARTIFICIAL INTELLIGENCE

A program that can sense, reason, act, and adapt.



MACHINE LEARNING

Algorithms whose performance improve as they are exposed to more data over time.



DEEP LEARNING

Subset of machine learning in which multilayered neural networks learn from vast amount of data.

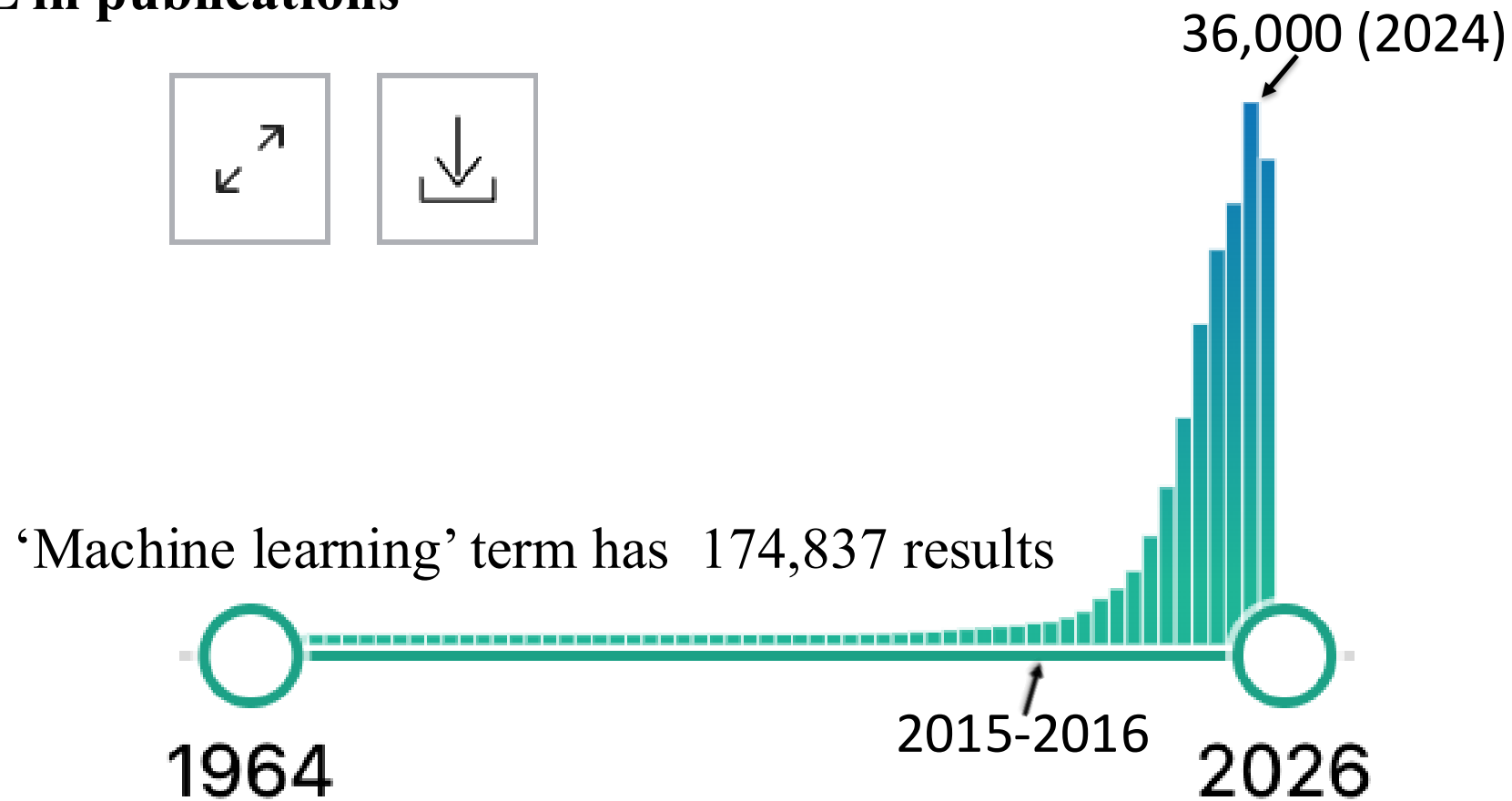
What is Machine learning ?

Machine Learning is the science of getting computers to act without being explicitly programmed (*Stanford University*).

Or

Machine Learning (ML) as a field within Artificial Intelligence (AI) focused on creating algorithms that learn from data to make predictions or decisions without explicit programming, enabling systems to improve automatically with experience.

Trend of ML in publications



<https://pubmed.ncbi.nlm.nih.gov/?term=%22Machine%20Learning%22>

Examples – Biological uses of AI



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SOUNDING BOARD | FROM THE NATIONAL ACADEMY OF MEDICINE



Generative AI in Medicine — Evaluating Progress and Challenges

Authors: Thomas M. Maddox, M.D., Peter Embí, M.D., Jackie Gerhart, M.D., Jennifer Goldsack, M.Chem., M.A., M.B.A., Ravi B. Parikh, M.D., M.P.P. , and Troy C. Sarich, Ph.D. [Author Info & Affiliations](#)

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AMERICAN JOURNAL OF PHYSIOLOGY

RENAL PHYSIOLOGY

Am J Physiol Renal Physiol 325: F395–F406, 2023.
First published August 17, 2023; doi:[10.1152/ajprenal.00177.2023](#)

REVIEW

“Hi, how can i help you?”: embracing artificial intelligence in kidney research

 **Anita T. Layton**^{1,2,3,4}

¹Department of Applied Mathematics, University of Waterloo, Waterloo, Ontario, Canada; ²Department of Biology, University of Waterloo, Waterloo, Ontario, Canada; ³Cheriton School of Computer Science, University of Waterloo, Waterloo, Ontario, Canada; and ⁴School of Pharmacology, University of Waterloo, Waterloo, Ontario, Canada

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Hypertension

REVIEW ARTICLE | Originally Published 21 February 2024 |  

 Check for updates

AI, Machine Learning, and ChatGPT in Hypertension

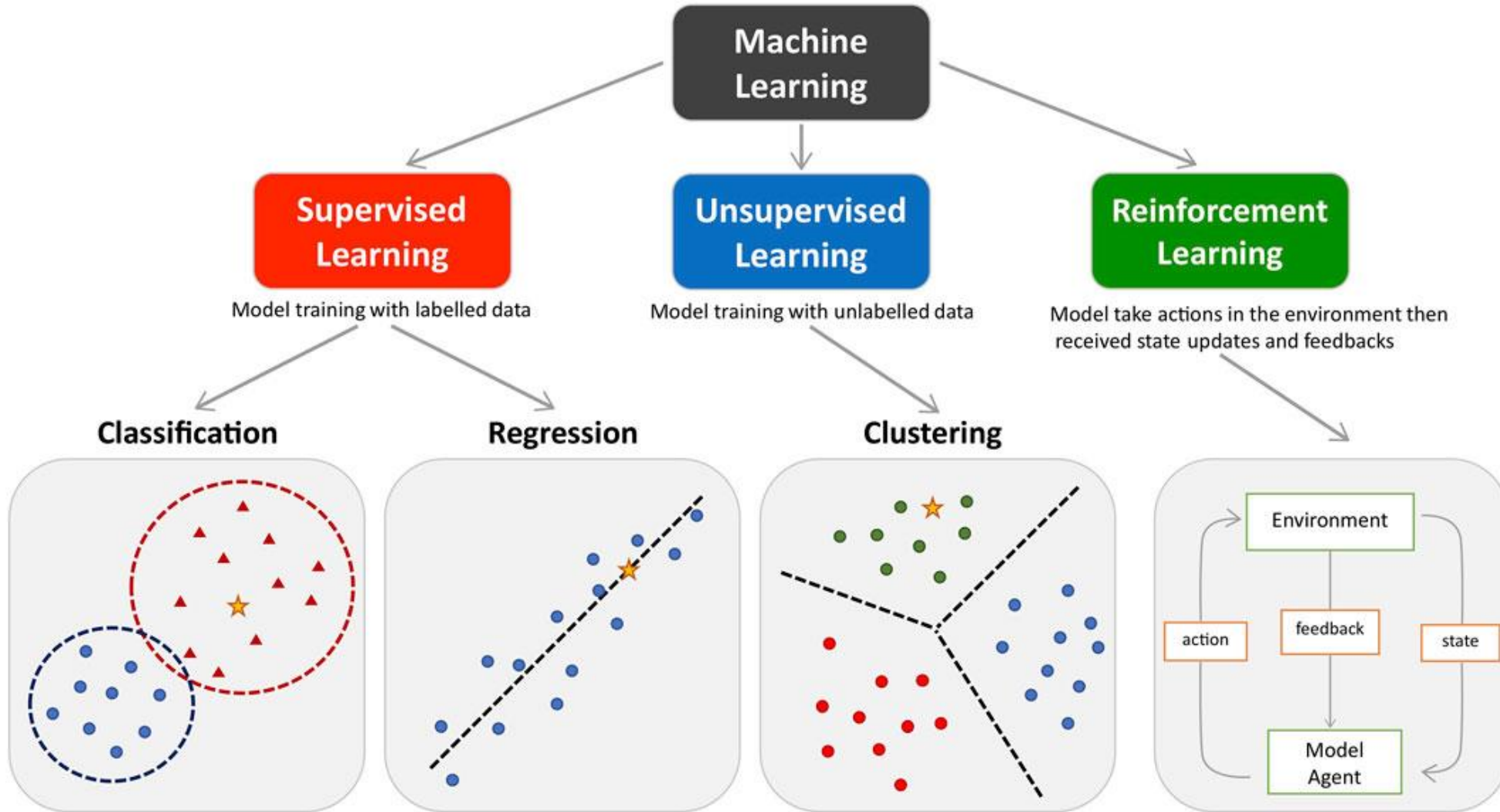
Anita T. Layton  | [AUTHOR INFO & AFFILIATIONS](#)

Hypertension • Volume 81, Number 4 • <https://doi.org/10.1161/HYPERTENSIONAHA.124.19468>

8,871 / 17



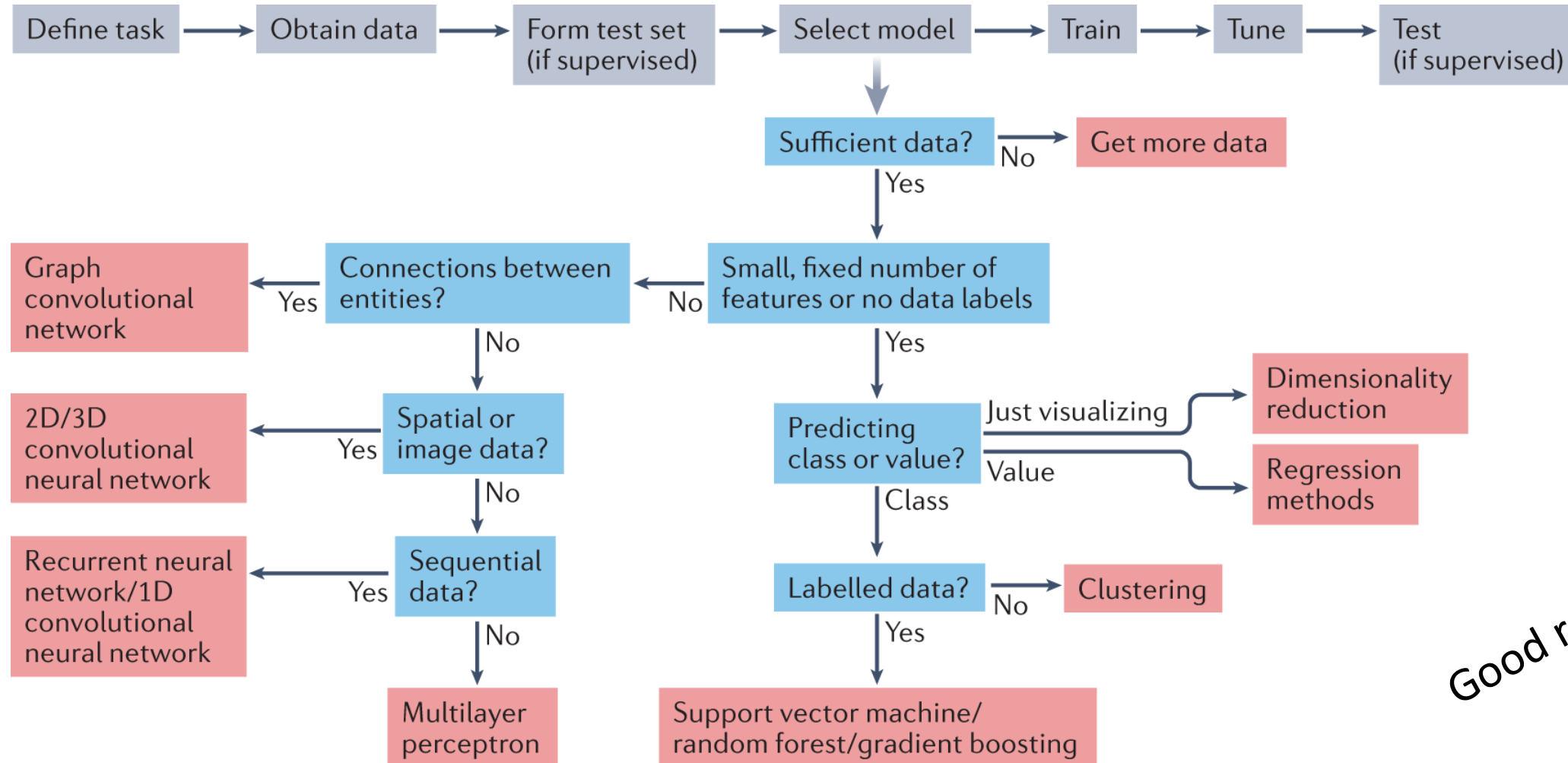
Sub-types of Machine learning ?



Workflow of machine learning study

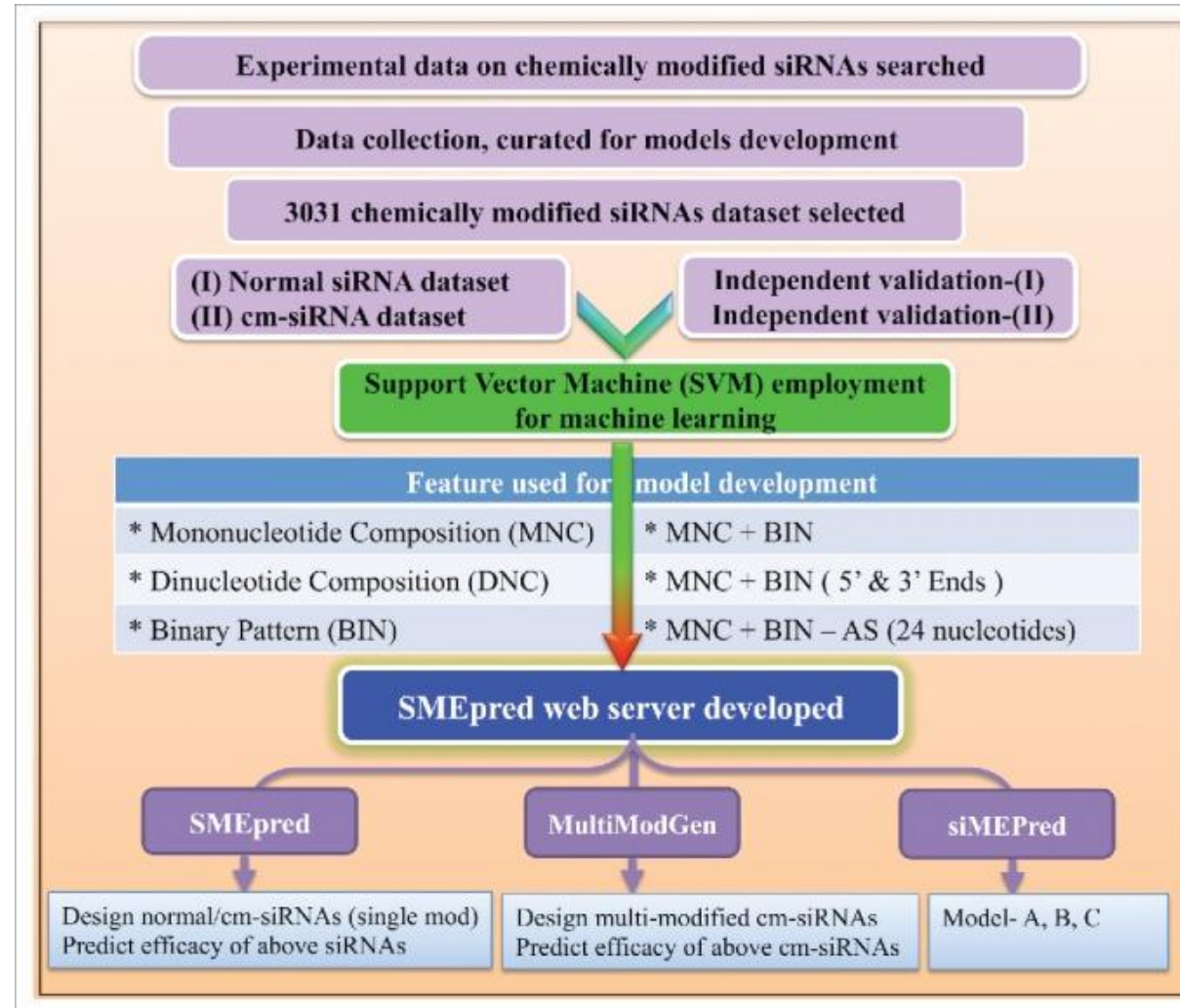


Choosing and training a ML method

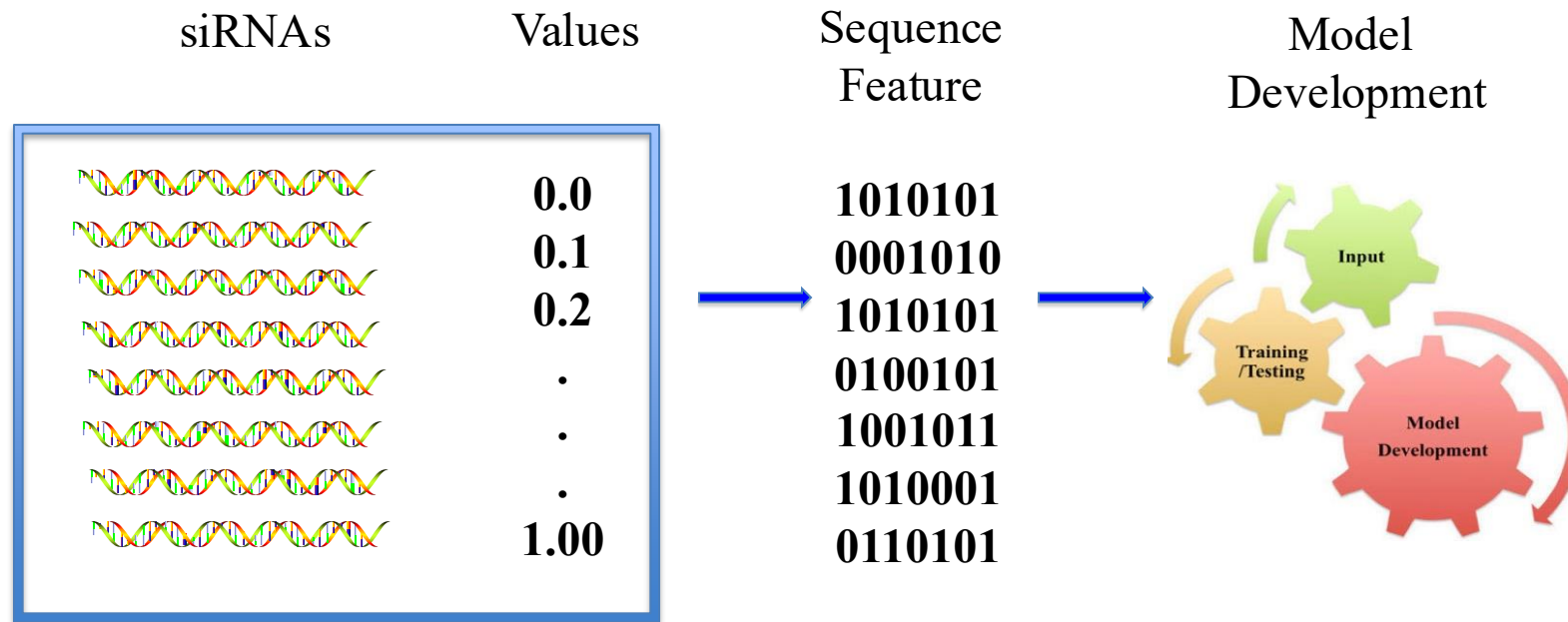


Good read !

Our workflow of hackathon.



Source: [SMEpred workbench: A web server for predicting efficacy of chemically modified siRNAs](#)

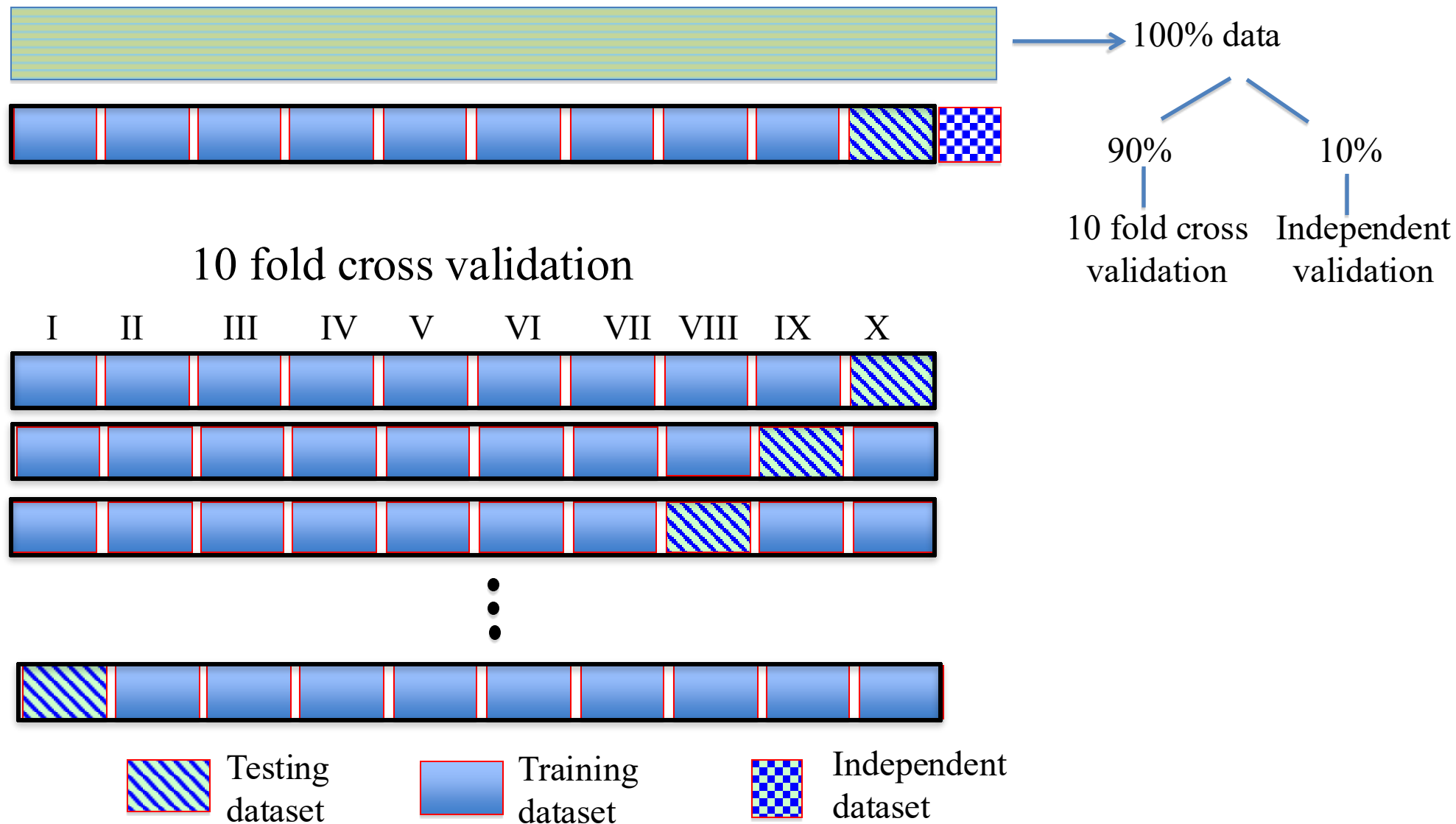


- Sequences - siRNAs (Modifications)
- ML/AI Models
 - I. Linear Regression (LR)
 - II. K-Nearest Neighbors (KNN)
 - III. Support Vector Machine (SVM)
 - IV. Artificial Neural Networks (ANN)
 - V. CNNs (1D)

Machine Learning & AI: The Basics

Evaluate a machine learning model's performance

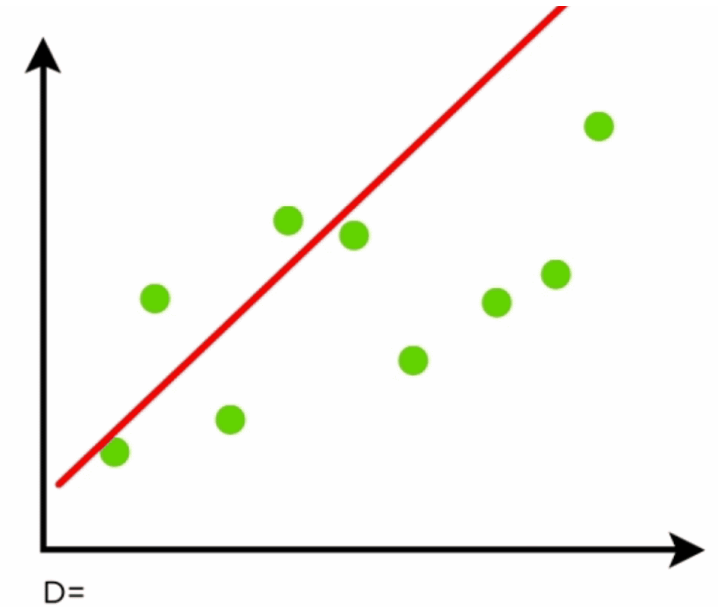
10 fold cross and independent validation



1) Linear Regression (LR)

Linear regression is a statistical technique used to model the linear relationship between a dependent variable and one or more independent variables.

The dependent variable is the variable that is being predicted, while the independent variables are the variables that are used to predict the dependent variable. Linear regression is used to find the equation of a line that best fits a set of data points.



Equation:

$$y' = b + w_1x_1$$

Diagram illustrating the components of the linear regression equation $y' = b + w_1x_1$:

- y' : Prediction
- b : Bias
- w_1 : Weight
- x_1 : Feature value

The Bias (b) and Weight (w_1) terms are calculated from training data.

$$Y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n + \varepsilon$$

Y is dependent variable(response/observation/y-variable)

β_0 is bias term

$\beta_1, \beta_2, \dots, \beta_n$ are parameter models(coefficient)

$x_1, x_2, \dots, x_n$ are independent variable(predictor/explanatory/x-variable)

ε is random error

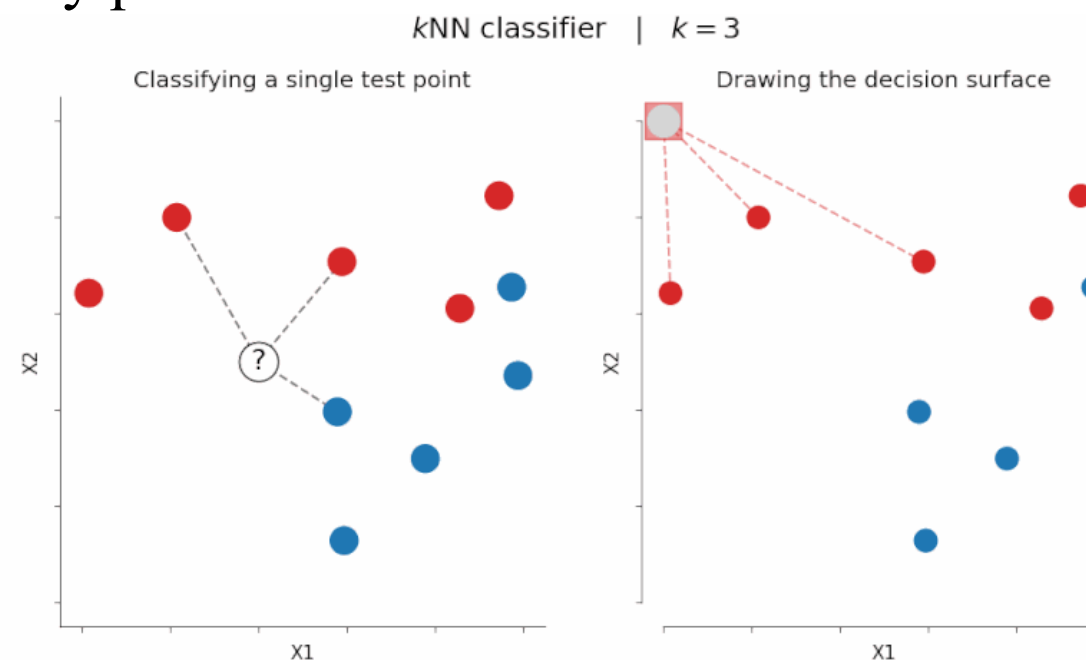
2) K-Nearest Neighbors (KNN)

K-Nearest Neighbors, is a simple, supervised machine learning algorithm that classifies new data points by finding the 'K' most similar known data points and determining the most frequent class among them.

The k value in the k-NN algorithm defines how many neighbors will be checked to determine the classification of a specific query point.

Euclidean distance (p=2)

$$d(x,y) = \sqrt{\sum_{i=1}^n (y_i - x_i)^2}$$



3) SVM- Support vector machines

A Support Vector Machine (SVM) is a supervised machine learning algorithm that finds the best possible boundary, or "hyperplane," to separate data points into different classes, with the goal of maximizing the "margin" (distance) between the hyperplane and the closest data points from each class, known as support vectors.

Equation for a hyperplane:

$$\mathbf{w} \cdot \mathbf{x} + \mathbf{b} = 0$$

In this equation:

(w) is the weight vector, representing the direction of the hyperplane.

(x) is the feature vector for a data point.

(b) is the bias term.

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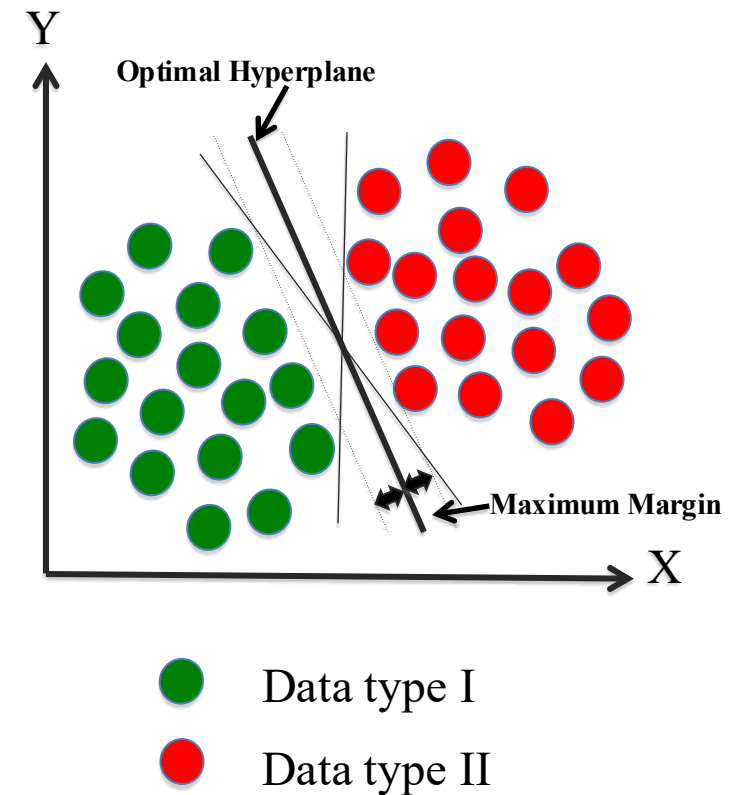
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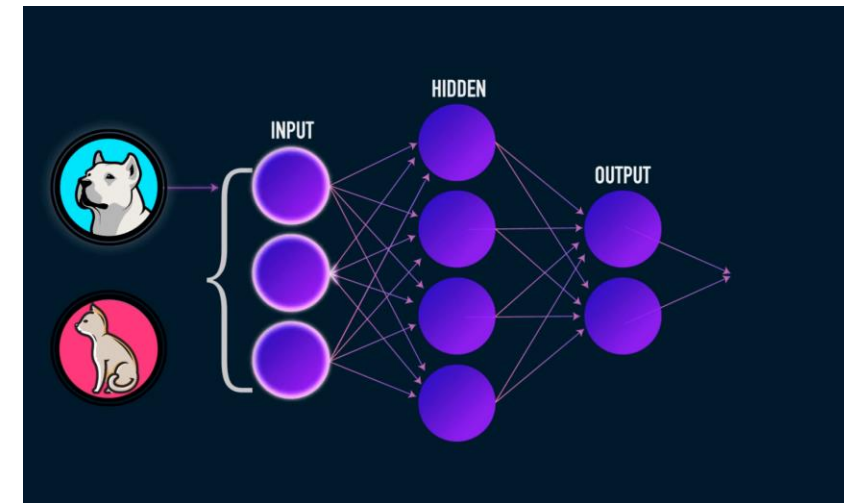
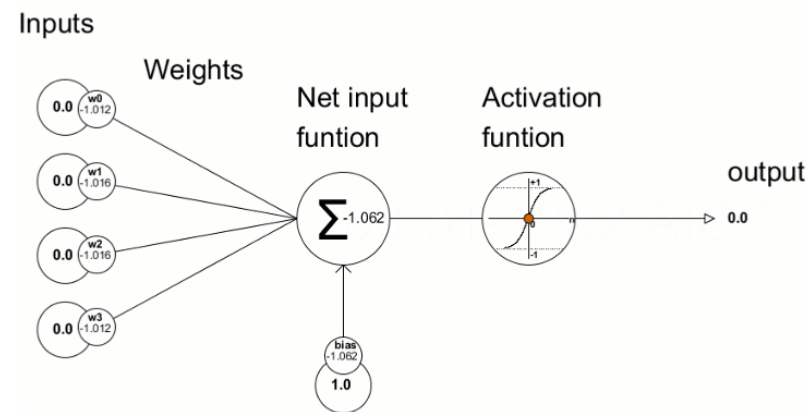
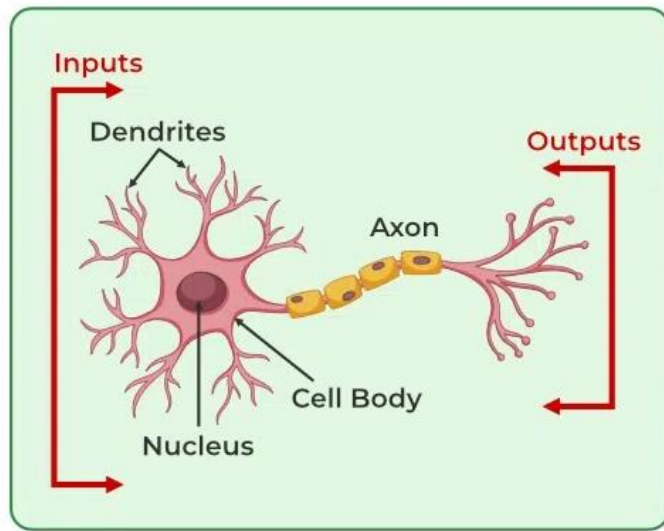
(x) is the feature vector for a data point.

(b) is the bias term.



4) Artificial Neural Network - ANN

An Artificial Neural Network (ANN) is a machine learning algorithm, inspired by the human brain's structure, that uses interconnected nodes ("neurons") to process information and learn patterns from data. Through a training process that adjusts the connection weights between neurons, an ANN learns to map input data to desired outputs, allowing it to make predictions or decisions on new, unseen data.



4) Artificial Neural Network - ANN

Aspect	Biological Neurons	Artificial Neurons
Structure	Dendrites: Receive signals from other neurons.	Input Nodes: Receive data and pass it on to the next layer.
	Cell Body (Soma): Processes the signals.	Hidden Layer Nodes: Process and transform the data.
	Axon: Transmits processed signals to other neurons.	Output Nodes: Produce the final result after processing.
Connections	Synapses: Links between neurons that transmit signals.	Weights: Connections between neurons that control the influence of one neuron on another.
Learning Mechanism	Synaptic Plasticity: Changes in synaptic strength based on activity over time.	Backpropagation: Adjusts the weights based on errors in predictions to improve future performance.
Activation	Activation: Neurons fire when signals are strong enough to reach a threshold.	Activation Function: Maps input to output, deciding if the neuron should fire based on the processed data.

4) Artificial Neural Network - ANN

- **Nodes and Layers:** ANNs are organized into layers of interconnected nodes (artificial neurons).
- **Input Layer:** Receives raw input data.
- **Hidden Layers:** Perform intermediate processing, transforming inputs into features.
- **Output Layer:** Produces the final prediction or output.
- **Weighted Connections:** Each connection between neurons has a weight, which determines the strength of the signal.
- **Activation Function:** Each neuron applies a mathematical activation function to its weighted inputs to determine its output.

Training

- **Initialization:** Connections start with random weights.
- **Input Data:** The network is presented with input data and a corresponding desired output.
- **Prediction:** The network processes the input and generates a predicted output.
- **Error Calculation:** The difference between the predicted and desired outputs is calculated.
- **Backpropagation/Gradient Descent:** An algorithm, often backpropagation, is used to adjust the connection weights to minimize the error.
- **Iteration:** This process is repeated many times with the training data until the network achieves a satisfactory level of accuracy.

5) Convolutional Neural Network - CNN

Convolutional Neural Network utilize specific layers (convolutional layers) to extract features, pooling layers to reduce data size, and fully connected layers to make final predictions to recognize patterns and classify information, and they are used in applications like image classification, natural language processing, and medical imaging.

Convolution is a mathematical operation that extracts features from input data, such as images, by using a filter (or kernel) that slides across the input.

A **1D convolutional layer** is a type of neural network layer that performs convolution operations on one-dimensional data. 2D convolutions that slide a kernel over a two-dimensional input (e.g., images).

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

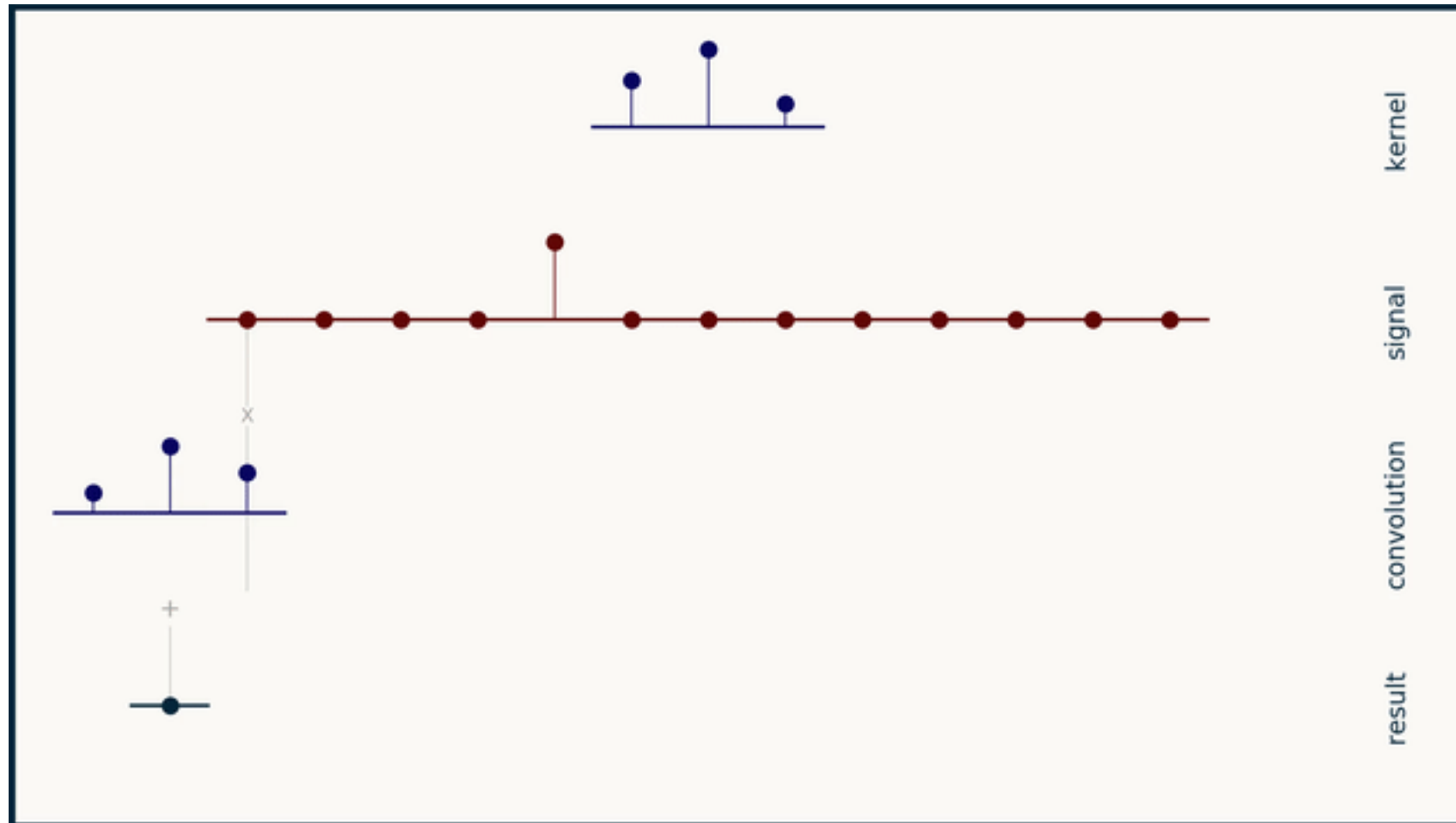
Image

4		

Convolved
Feature

5) Convolutional Neural Network-CNN

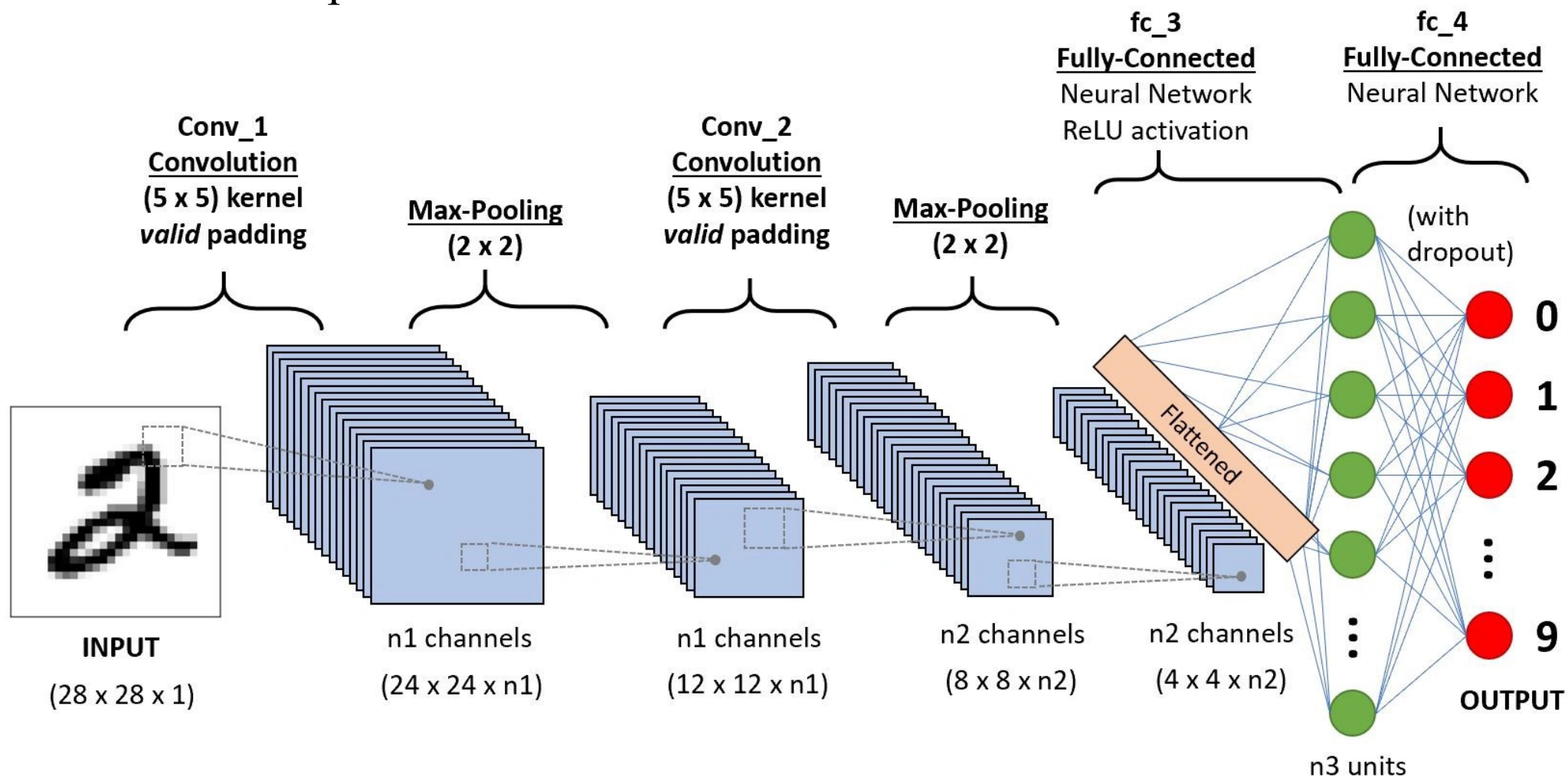
Convolution in one dimension for neural networks



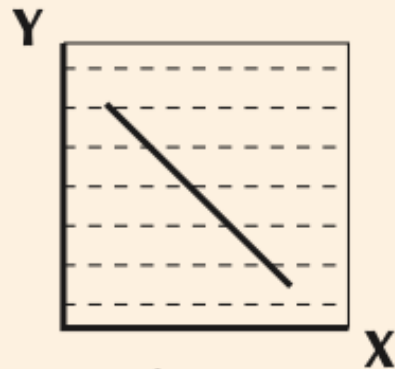
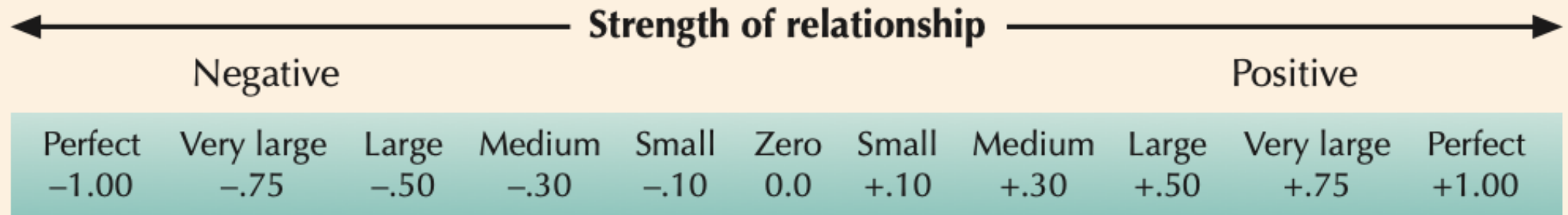
Source https://www.brandonrohrer.com/convolution_one_d

5) Convolutional Neural Network-CNN

Deeper look

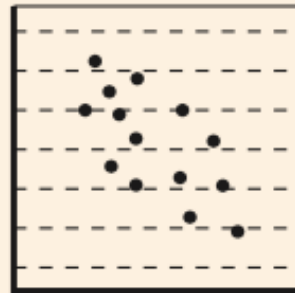


Correlation – relationship b/w actual and predicted values



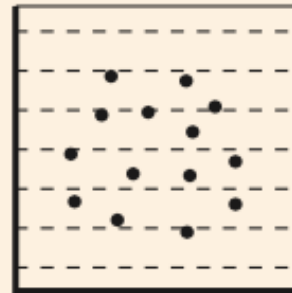
Perfect
negative
relationship

(a)



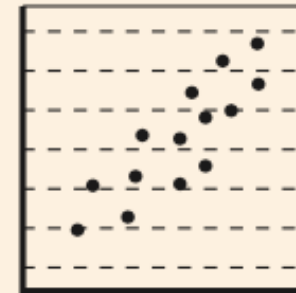
Moderate
negative
relationship

(b)



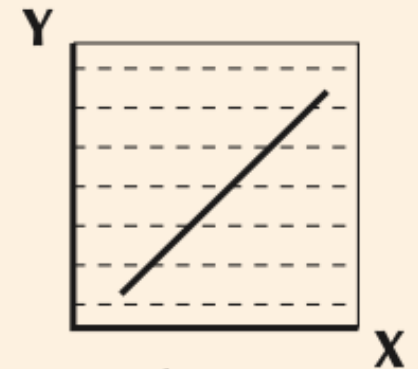
No relationship

(c)



Moderate
positive
relationship

(d)



Perfect
positive
relationship

(e)

Calculate Pearson correlation

x_i are the **individual values** of one **variable** e.g. age

y_i are the **individual values** of the other **variable** e.g. salary

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where r is the Pearson correlation coefficient,

$\bar{x}$ and $\bar{y}$ are respectively the **mean values** of the two variables.

Step-1

- Data collection, cleaning & curation

Step-2

- Feature selection of the dataset

Step-3

- Model Development - 10-fold cross & independent validation

Step-4

- Machine Learning Models

Step-5

- Comparisons



Showkat Ahmad Dar
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Additional Papers ...

Cm-siRPred: Predicting chemically modified siRNA efficiency based on multi-view learning strategy

Tianyuan Liu ^a, Junyang Huang ^b, Delun Luo ^{a,c}, Liping Ren ^d, Lin Ning ^d, Jian Huang ^e , Hao Lin ^e , Yang Zhang ^a 

